SESSION 6 – Probability Basics

Task 1

Question:
List all possible outcomes (sample space) for rolling a standard six-sided die and flipping a coin at the same time. Write your answer as a list of ordered pairs.

Answer:

A die has outcomes {1, 2, 3, 4, 5, 6} and a coin has outcomes {H, T}.

The sample space is:

S = {
(1,H), (1,T),
(2,H), (2,T),
(3,H), (3,T),
(4,H), (4,T),
(5,H), (5,T),
(6,H), (6,T)
}

Total number of outcomes = 6 × 2 = 12

Task 2

Question:
Suppose you have a deck of 10 cards numbered 1 to 10. Calculate the classical probability of drawing a card with an even number. Show your calculation steps.

Answer:

Cards are:

```
1, 2, 3, 4, 5, 6, 7, 8, 9, 10
```

Even-numbered cards are:

```
2, 4, 6, 8, 10
```

Number of favorable outcomes = **5**

Total number of outcomes = **10**

Using the classical probability formula:

P(Even) = Favorable outcomes / Total outcomes
P(Even) = 5 / 10
= 0.5

Final Answer: P(Even) = 0.5 or 50%

Task 3

Question:
In a playlist of 8 songs on Spotify, 3 are by Arijit Singh. If you shuffle and play one random song, what is the probability that it is NOT by Arijit Singh? Show your working.

Answer:

Total songs = 8

Songs by Arijit Singh = 3

Songs NOT by Arijit Singh:

8 - 3 = 5

Therefore:

P(Not Arijit Singh) = Favorable outcomes / Total outcomes

= 5 / 8

= 0.625

As a percentage:

0.625 × 100 = 62.5%

Final Answer: P(Not Arijit Singh) = 0.625 or 62.5%

Task 4

Question:
A Zomato user places an order. The probability they order Pizza is 0.3, and the probability they order Burger is 0.4. If the probability they order both is 0.1, use the addition rule to find the probability the user orders Pizza or Burger.

Answer:

P(Pizza) = 0.3
P(Burger) = 0.4
P(Pizza and Burger) = 0.1

Using the addition rule:

P(Pizza or Burger)
= P(Pizza) + P(Burger) - P(Pizza and Burger)

= 0.3 + 0.4 - 0.1

= 0.6

Final Answer: P(Pizza or Burger) = 0.6 or 60%

Task 5

Question:
A Flipkart user is browsing electronics. The probability they add a phone to cart is 0.2. If you know they are interested in mobiles, the probability rises to 0.5. Briefly explain in your own words what this conditional probability means, using this scenario.

Answer:

The conditional probability means that the probability of a user adding a phone to their cart **changes when we know additional information about the user**.

Normally:

P(Add phone) = 0.2 = 20%

But if we already know that the user is interested in mobiles:

P(Add phone | Interested in mobiles) = 0.5 = 50%

In simple words:
Knowing that the user is interested in mobiles increases the probability of them adding a phone to the cart from **20% to 50%**.

Final Answer: Conditional probability tells us the probability of an event when we already know that another condition or event has occurred.